DA3408 Assignment 1 Writeup

Question 1: Technical Debt Diagnosis

(a) Entanglement (CACE). Changing the delivery-time feature also changed an unrelated recommendation feature. This shows how one change can affect other parts of an ML system.

(b) Undeclared Consumers. The marketing team used the model's raw output table without telling the ML team. The ML team therefore did not know about this dependency when changing the model or its output format.

(c) Configuration Debt and Glue Code/Pipeline Jungles. The fourteen undocumented shell scripts form a pipeline that is hard to understand and maintain. There is no clear orchestration or record of the dependencies.

For (b), marketing should stop reading the raw output table directly. The ML team should provide a documented, versioned interface and list marketing as a known consumer. The approved model can be accessed through the MLflow Model Registry using a stage such as `Production`. A model signature can also be logged to check the expected inputs and outputs before a change is deployed. This makes the dependency visible to both teams.

Question 2: MLflow Experiment Analysis

I ran six MLP experiments on the same MNIST split by combining learning rates `0.001`, `0.003`, and `0.01` with batch sizes `32` and `128`. I kept the model, seed, and number of epochs fixed. The best run was `mlp-lr-0.001-bs-32`. Its best and final validation accuracy was `0.9735`, the highest among the six runs.

The best run did not show clear overfitting. Its validation accuracy reached its highest value at the final epoch while its training loss kept decreasing. Some other settings were less stable. For example, `lr=0.003, bs=128` reached `0.9703` but ended at `0.9676`, even though the training loss continued to fall.

Learning rate had a larger effect than batch size. The average best validation accuracy was about `0.9728`, `0.9696`, and `0.9487` for learning rates `0.001`, `0.003`, and `0.01`, a range of about 2.41 percentage points. The two batch-size averages differed by only about 0.79 percentage points. Thus, learning rate had the greater effect in these experiments. More runs with different seeds would make this conclusion more reliable.

Question 3: DVC Versioning

DVC stored two dataset versions in an SSH directory on Aakash's Ubuntu VM: `ssh://aakash@10.22.18.20:2222/home/aakash/dvcstore/q3-arnav`. Version `q3-v1` contained 1800 filenames and version `q3-v2` contained 2800. Running `git checkout q3-v1` followed by `dvc checkout` restored the first version, and `wc -l` showed 1801 lines including the CSV header. Returning to `main` and running `dvc checkout` restored 2801 lines. This confirmed that Git and DVC could restore the older dataset and then return to the latest one.

Question 4: Partner Reproduction

Aakash (Partner A) ran the original experiment. I (Partner B) recreated his Conda environment and reran the experiment with identical parameters. Both Aakash's run and my run produced validation accuracy `0.948` and training loss `0.02032236113371596`. The accuracy difference was `0.000`, within the chosen tolerance of `0.001`, so the reproduction was a match. I committed my reproduction evidence to Aakash's repository as a separate contribution.